

Warm-up problem

2. Goats walk around with very large abdomens while mountain lions, in contrast, do not. Can you propose a reason and support it with some quantitative estimates?



Well, let's try to think of each animal as a reactor that eats food, extracts chemical energy, and excretes waste. And let's assume that each animal requires about the same amount of extracted chemical energy per kg of body weight. So then what is the quantity of food they consume? On average, we might expect:

$$\text{Consumption rate} = \text{energy need} \cdot \frac{1}{\text{energy recovery of food}}$$

$(\text{kg}_{\text{food}} \text{ kg}_{\text{body}}^{-1} \text{ d}^{-1})$ $(\text{MJ kg}_{\text{body}}^{-1} \text{ d}^{-1})$ $(\text{MJ kg}_{\text{food}}^{-1})$

Is the energy content of mountain lion food different from goat food? Maybe not. Goats are grazers, and consume grass and other plants, which are primarily carbohydrate. Mountain lions are predators, and eat mostly protein, which has a similar energy content, but presumably also some fat, which does have a much higher energy content. How much fat though? I don't know.

In contrast, the digestability of the different diets is undoubtedly different. Much less of the total energy content is actually recovered from the plant-based goat diet. So the "energy recovery" term above is smaller for goats, and therefore if our equation is correct, they must consume more food.

Probably more important, the degradable part of goat food is degraded more slowly than protein, which can be rapidly hydrolyzed by enzymes. Plant fiber, in contrast, takes an awful lot of chewing and digesting, even with help from gut microorganisms. Goats, like all ruminants (including cattle) have four stomach chambers to help, and regurgitate their food to chew it multiple times.

Thinking about a simple first-order model,

$$\begin{array}{c} \dot{E}_{\text{ext.}} \\ \text{Rate of energy} \\ \text{recovery} \end{array} = \begin{array}{c} M_{\text{food}} \\ \text{mass of} \\ \text{food present} \end{array} \cdot k \quad \begin{array}{c} \text{Rate constant} \\ k \end{array}$$

$$\left(\text{MJ kg}^{-1} \text{d}^{-1} \right) \quad \left(\text{kg kg}^{-1} \right) \quad \left(\text{d}^{-1} \right)$$

If the rate of energy recovery is similar but the rate constant k is smaller for goats, the mass of food present in the "reactor" must be larger. So that is one answer.

To get quantitative, we are really going out on a limb. But perhaps we can assume that k for goat food is less than half of k for lion food (probably much less, so this is a safe guess). Then from the equation above, we would estimate that goats carry around twice as much food on average.

Now, there are two different ways that could happen: by a higher consumption or higher retention time. If you are interested in this distinction, read on.

To go further, let's treat the animals as plug-flow reactors and think about daily food consumption flowing through. Then

$$\begin{array}{c} \dot{E}_{\text{ext.}} \\ \text{Rate of energy} \\ \text{extraction} \end{array} = (1 - e^{-k \cdot T_s}) \begin{array}{c} \dot{m}_{\text{food}} \\ \text{mass flow} \\ \text{rate of food} \end{array} \cdot \begin{array}{c} e_{\text{chem}} \\ \text{Extractable energy} \\ \text{in food} \end{array}$$

$$\left(\text{MJ kg}^{-1} \text{d}^{-1} \right) \quad \left(\text{kg kg}^{-1} \text{d}^{-1} \right) \quad \left(\text{MJ kg}^{-1} \right)$$

$T_s = \text{solids retention time. (d)}$

So if k is smaller for goats, we have two options to get the same rate of energy extraction: 1) increase mass flow rate of food, i.e., food consumption rate, while keeping solids retention time constant or 2) increase retention time for a fixed consumption rate. Of course some combination is possible too. But in any case,

$$\tau = \frac{M_{\text{food}}}{\dot{m}_{\text{food}}} \rightarrow M_{\text{food}} = \dot{m}_{\text{food}} \cdot \tau$$

the mass of food in the "reactor" must be larger if k is lower. How much larger? Using our factor of two guess above, if we keep consumption the same then retention time must be exactly twice as large to get the same result. So the total mass of digestible food present would be twice as much. The first option, increasing consumption, could also be solved.